

MAINCRAFTS TECHNOLOGIES

Cybersecurity Task-1

Threat Report (Awareness & Research Project)

Name: Purva Chandrashekhar Chatare

M.Sc. Cybersecurity

Amity University Rajasthan, Jaipur.

Table of Contents

- 1. Introduction to Cybersecurity**
 - a. 1.1 What is Cybersecurity?
 - b. 1.2 Importance & Current Relevance
- 2. Threat Landscape: Top 5 Modern Cyber Threats**
 - a. 2.1 AI-Powered Phishing & Deepfakes
 - b. 2.2 Ransomware-as-a-Service (RaaS)
 - c. 2.3 Cloud Security Misconfigurations
 - d. 2.4 IoT Vulnerabilities (Mirai Botnets)
 - e. 2.5 Zero-Day Exploits (Supply Chain)
- 3. Impact Analysis (Individuals vs. Organizations)**
- 4. Real-World Case Studies (2024–2025)**
- 5. Preventive Measures & Strategic Defense**
- 6. Conclusion & Future Outlook**
- 7. References**

1. Introduction to Cybersecurity

1.1 What is Cybersecurity?

Cybersecurity is the practice of protecting systems, networks, programs, and data from digital attacks, damage, or unauthorized access. It encompasses a range of disciplines, including information security, network defense, disaster recovery, and end-user education.

1.2 Why is it Important?

For **individuals**, cybersecurity is crucial to safeguard financial assets, privacy, and identity against fraud. For **businesses**, it is a cornerstone of operational resilience. A breach can result in devastating financial losses, legal liabilities (GDPR, CCPA), and irreparable brand damage.

1.3 Current Relevance (The 2025 Landscape)

As of 2025, the digital dependency has reached critical mass, but so have the threats. Attackers are no longer just hackers; they are state-sponsored entities and organized crime rings utilizing Generative AI. The rise of **AI-driven attacks** and **Ransomware-as-a-Service (RaaS)** has lowered the technical barrier for criminals, making cybercrime a pervasive reality for critical infrastructure and private citizens alike .

2. Threat Analysis: The Big 5 (2024–2025)

2.1 AI-Powered Phishing & Deepfakes

Traditional phishing relied on spelling errors. Modern phishing uses Generative AI to craft perfect, context-aware emails without grammatical mistakes. Attackers use AI to generate realistic "deepfake" audio/video for social engineering and obfuscated code (e.g., SVG attachments) that evades traditional scanners .

2.2 Ransomware-as-a-Service (RaaS)

RaaS has democratized ransomware. Affiliates pay to use ransomware infrastructure developed by core groups (like the now-disrupted LockBit or the evolved RansomHub/DragonForce). These groups use "double extortion"—encrypting data and threatening to leak it publicly .

2.3 Cloud Misconfigurations

The #1 cause of cloud data breaches remains human error. Developers often leave storage containers (e.g., AWS S3 buckets) publicly accessible or hard-

code API keys into source code. This leads to massive data leaks of Personally Identifiable Information (PII) .

2.4 IoT Vulnerabilities

Smart devices (DVRs, cameras, routers) are notoriously insecure due to default passwords and lack of patching. Botnets like **Mirai** (still active in 2025) hijack these devices to launch massive Distributed Denial of Service (DDoS) attacks that can take down entire national internet infrastructures .

2.5 Zero-Day Exploits

These are previously unknown vulnerabilities in software. Nation-state actors (like Volt Typhoon) weaponize these flaws before vendors can issue a patch, targeting network management systems to establish long-term persistence for espionage .

3. Impact Analysis

Threat Vector	Impact on Individuals	Impact on Organizations
AI Phishing	Identity theft, drained bank accounts via fake login pages.	Credential compromise of C-suite executives, data breaches.
RaaS	Loss of personal files/photos if personal devices are hit.	Operational downtime (weeks), multi-million dollar ransom demands, stock price drop.
Cloud Misconfig	Exposure of personal medical/financial records on public internet.	Regulatory fines (e.g., \$370k+ average breach cost), loss of customer trust .
IoT Botnets	Privacy invasion (camera feeds exposed). Device used in attacks.	Critical infrastructure disruption (power grids, factories). DDoS downtime.

Zero-Day	Malware installation via drive-by downloads.	Complete supply chain compromise (SolarWinds-style).
-----------------	--	--

4. Real-World Case Studies (2024–2025)

1. AI-Powered Phishing: The SVG Obfuscation Campaign (2025)

Incident: In August 2025, Microsoft Threat Intelligence detected a campaign using **AI-generated SVG files** disguised as PDFs. The code used business jargon (e.g., "revenue," "risk") as variables to evade detection. The AI-generated code was so verbose it became a signature for detection .

Defense Used: AI-powered defense (Microsoft Security Copilot) analyzed behavioral cues rather than just code signatures.

2. RaaS: The Rise and Fall of LockBit & RansomHub (2024-2025)

Incident: LockBit, once the most prolific RaaS, was disrupted by "Operation Cronos" in 2024. However, in 2024-2025, **RansomHub** emerged, breaching over 600 organizations by targeting ESXi virtual machines. When RansomHub disappeared, affiliates immediately rebranded to **DragonForce**, proving the resilience of the RaaS model .

3. Cloud Misconfiguration: Football Australia Breach (2024)

Incident: Researchers found plaintext AWS keys embedded directly in the source code of Football Australia’s website. This misconfiguration exposed 127 digital storage containers containing player personal data for over 700 days .

4. IoT Vulnerabilities: New Mirai Variant (2025)

Incident: Kaspersky honeypots detected a new Mirai variant targeting DVR devices. The malware specifically checked if it was running in a virtual machine (VM) to evade analysis by security researchers, showcasing anti-detection tactics .

5. Zero-Day: Versa Director Exploitation (2024-2025)

Incident: State-aligned groups (VOLT Typhoon) exploited CVE-2024-39717, a zero-day in Versa Director (SD-WAN management). Attackers deployed a custom web shell "VersaMem" to harvest credentials from ISPs and manage espionage operations .

5. Preventive Measures (Defense in Depth)

To counter these threats, organizations must move beyond traditional perimeter security. Below is a matrix of specific solutions mapped to the threats.

5.1 AI-Powered Phishing Attacks

No.	Measure Type	Solution
1	Prevent	Deploy Phishing-resistant MFA (FIDO2 security keys)
2	Detect	Use AI-driven email filtering (behavioral analysis)
3	Respond	Conduct Security Awareness Training with deepfake simulations

5.2 Ransomware-as-a-Service (RaaS)

No.	Measure Type	Solution
1	Prevent	Implement Zero Trust Architecture (never trust, always verify)
2	Detect	Deploy Endpoint Detection and Response (EDR) for anomaly monitoring
3	Respond	Maintain Offline, Immutable Backups (air-gapped storage)

5.3 Cloud Security Misconfigurations

No.	Measure Type	Solution
1	Prevent	Use Infrastructure as Code (IaC) for automated secure configurations
2	Detect	Deploy Cloud Security Posture Management (CSPM) tools
3	Respond	Apply Principle of Least Privilege with short-term credentials (OIDC)

5.4 IoT Vulnerabilities

No.	Measure Type	Solution
1	Prevent	Change default credentials and implement Network Segmentation (VLANs)
2	Detect	Set up Honeypots and monitor unusual outbound traffic
3	Respond	Perform Regular Firmware Patch Management

5.5 Zero-Day Exploits

No.	Measure Type	Solution
1	Prevent	Conduct Attack Surface Management (reduce exposed ports)
2	Detect	Deploy Runtime Application Self-Protection (RASP)
3	Respond	Use Virtual Patching (WAF rules before vendor patch is ready)

6. Conclusion & Future Scope

The threat landscape of 2025 is defined by the **commoditization of AI** for both attackers and defenders. The old model of "trust but verify" is dead; we are moving entirely to "**Zero Trust.**"

For a business to survive, cybersecurity cannot be an afterthought—it must be embedded in the development lifecycle (DevSecOps). The RaaS cases of LockBit and RansomHub prove that while law enforcement can disrupt infrastructure, the **business model of cybercrime adapts instantly** .

Future Focus: We must prepare for **AI vs. AI** battles where defensive AI systems fight autonomous offensive AI agents. Continuous learning, threat hunting, and sharing intelligence (via CISA alerts) are the only ways to stay ahead .

7. References

1. Microsoft. (2025). *AI-powered defences against novel SVG phishing*.
2. TU Delft. (2025). *Empirical Analysis of LockBit's Business Operations*.
3. Cloud Security Alliance. (2025). *2024 Football Australia Data Breach Case Study*.
4. Kaspersky. (2025). *Mirai Botnet targeting IoT Devices*.
5. SecPod. (2025). *Versa Director Zero-Day Exploitation*.
6. CISA / PYA. (2025). *Nation-State Cyber Threat Advisory*.
7. OWASP. (2025). **Non-Human Identities Top 10 (NHI6:2025)**.
8. Cloud Security Alliance. (2025). *RansomHub ESXi Tactics*.